

IL'INSKOYE, Russian Federation

WMO: 321210

Lat: 47.9867N

Lon: 142.2050E

Elev: 17

StdP: 101.12

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.1	(c) -18.7	(d) -24.5	(e) 0.4	(f) -20.2	(g) -22.4	(h) 0.5	(i) -18.0	(j) 15.0	(k) -7.7	(l) 13.8	(m) -8.7	(n) 2.0	(o) 110	(p) 0.713

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.0	(c) 22.2	(d) 19.8	(e) 21.2	(f) 19.0	(g) 20.2	(h) 18.2	(i) 20.3	(j) 21.6	(k) 19.4	(l) 20.8	(m) 18.5	(n) 20.0	(o) 4.8	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 19.7	(b) 14.5	(c) 21.2	(d) 18.9	(e) 13.7	(f) 20.5	(g) 17.9	(h) 12.9	(i) 19.7	(j) 58.8	(k) 21.8	(l) 55.8	(m) 20.8	(n) 52.9	(o) 20.0	(p) 23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.5	11.7	10.3	-26.4	24.2	1.9	1.3	-27.8	25.1	-28.9	25.9	-30.0	26.6	-31.4	27.5
			-26.6	21.2	1.9	1.1	-28.0	22.0	-29.0	22.6	-30.1	23.2	-31.5	24.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-10.8	-10.4	-4.8	1.3	6.2	10.8	15.0	17.0	13.9	7.2	-0.9	-8.0
	DBStd	10.25	4.30	4.47	3.85	2.81	3.05	2.68	2.48	2.41	2.86	3.37	4.41	4.18
	HDD10.0	3068	645	571	457	260	125	22	1	0	4	99	327	557
	HDD18.3	5568	903	804	716	509	377	227	104	55	135	345	577	816
	CDD10.0	559	0	0	0	0	6	45	157	218	120	13	0	0
	CDD18.3	18	0	0	0	0	0	0	2	15	1	0	0	0
	CDH23.3	6	0	0	0	0	0	1	1	4	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.4	4.6	4.3	4.6	4.5	4.4	3.8	3.6	3.7	4.2	5.0	5.4	5.0
Precipitation	PrecAvg	829	57	42	46	51	64	54	85	101	92	94	73	70
	PrecMax	1170	103	85	105	114	123	109	271	212	173	191	157	129
	PrecMin	634	18	2	17	14	14	3	5	28	32	44	21	15
	PrecStd	138	18	19	25	25	26	29	64	43	34	33	36	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.5	1.7	4.6	11.5	16.4	20.5	22.2	23.9	22.2	17.2	10.9	3.8
		MCWB	-0.3	0.5	2.6	7.1	11.6	16.2	19.0	21.2	19.2	14.6	9.5	3.1
	2%	DB	-1.2	-0.3	3.0	8.8	14.2	17.4	20.8	22.7	20.7	15.4	8.8	1.7
		MCWB	-2.4	-1.4	1.2	5.8	10.5	14.3	18.5	20.5	18.1	12.9	7.3	0.6
	5%	DB	-2.7	-1.8	1.7	7.3	12.6	16.0	19.9	21.9	19.7	14.0	7.1	-0.1
		MCWB	-3.8	-2.8	0.3	4.9	9.5	13.8	17.9	19.8	17.3	11.8	5.6	-1.2
	10%	DB	-4.2	-3.5	0.6	5.9	11.2	14.8	19.0	21.0	18.6	12.6	5.5	-1.5
		MCWB	-5.3	-4.7	-0.8	3.8	8.7	12.9	17.3	19.0	16.3	10.4	3.9	-2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.3	0.8	3.1	7.7	12.5	16.5	19.9	22.1	19.9	15.0	9.7	3.2
		MCDB	0.2	1.6	4.1	10.2	15.1	19.8	21.2	23.2	21.5	16.5	10.8	3.7
	2%	WB	-2.3	-1.2	1.5	6.2	11.1	15.0	19.0	21.1	18.6	13.3	7.4	0.7
		MCDB	-1.4	-0.4	2.6	8.3	13.4	16.7	20.3	22.3	20.2	15.0	8.7	1.5
	5%	WB	-3.7	-2.8	0.4	5.1	10.0	14.1	18.3	20.2	17.7	12.0	5.7	-1.1
		MCDB	-2.7	-1.8	1.6	7.1	12.1	15.6	19.6	21.4	19.4	13.6	6.9	-0.2
	10%	WB	-5.2	-4.6	-0.7	3.9	8.9	13.1	17.5	19.4	16.6	10.7	4.1	-2.6
		MCDB	-4.3	-3.6	0.5	5.7	10.9	14.5	18.8	20.8	18.3	12.4	5.4	-1.6
Mean Daily Temperature Range	5% DB	MDBR	6.3	7.3	6.0	5.0	5.4	5.2	4.7	5.0	6.4	6.3	5.1	5.7
		MCDBR	6.7	7.5	6.7	7.5	7.5	7.3	6.0	5.7	7.4	7.5	6.5	6.5
		MCWBR	6.4	6.9	5.8	5.5	5.0	5.0	4.2	4.3	5.7	6.5	6.2	6.4
	5% WB	MCDBR	6.4	6.9	6.3	7.0	6.6	6.5	5.3	4.8	6.6	7.2	6.5	6.5
		MCWBR	6.2	6.6	5.8	5.5	4.7	4.7	4.1	3.9	5.6	6.6	6.4	6.5
Clear-Sky Solar Irradiance	taub	0.316	0.353	0.412	0.498	0.531	0.501	0.464	0.440	0.397	0.404	0.377	0.338	
	taud	2.099	1.974	1.892	1.812	1.840	2.036	2.219	2.298	2.344	2.145	2.102	2.083	
	Ebn at Noon	716	766	779	750	745	774	799	803	804	709	630	626	
	Edh at Noon	89	128	164	197	199	164	135	120	104	108	91	82	
All-Sky Solar Radiation	RadAvg	1.28	2.24	3.42	4.43	4.77	4.96	4.48	4.24	3.73	2.44	1.35	0.94	
	RadStd	0.14	0.20	0.33	0.48	0.46	0.51	0.50	0.40	0.35	0.16	0.12	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page